

Granite Heron Mitigation Expert Report

Mitigation and economic-loss opinions for Granite Heron Logistics, Inc.

Civil Action No. SYN-WDNC-24-CV-0520

Report dated August 4, 2025

Mercer Webb PLLC retained me to analyze whether Blue Ember Biologics, LLC had an economically reasonable opportunity to reduce delay after rejecting validation lot BE214-VL-2409A. I also evaluated the structure of Blue Ember's 4.2 million dollar claim. I do not offer a carrier-operation, product-stability, regulatory, or legal-preservation opinion.

My central opinion is that the October 2, 2024 Cobalt Finch offer presented a commercially reasonable accelerated alternative. Had Blue Ember secured the December 2 slot and performed the proposed transfer, it would have avoided 1.8 million dollars of the claimed delay at approximately 300,000 dollars of incremental transfer and qualification expense. The net avoidable loss is 1.5 million dollars.

For this mitigation calculation, I accept Blue Ember's 4.2 million gross model as a conditional starting point, not as an admission that each component is recoverable. Applying the net deduction produces 2.7 million dollars. Liability, original-lot cost, and replacement expense remain analytically distinct from the post-event capacity decision.

Professional background and payment

I hold fictional doctoral and master's degrees in operations economics and have twenty years of experience analyzing constrained manufacturing, technology transfer, project delay, expected cost, and loss reduction. My consulting work has included development-stage life sciences, specialty chemicals, and high-reliability supply chains.

I use schedule networks, contemporaneous capacity evidence, incremental-cost comparison, probability assessment, and sensitivity testing. I have published fictional articles on decision-making under compressed qualification timelines and taught graduate courses addressing replacement capacity after a disruption.

My billing rate is 575 dollars per hour for analysis and 700 dollars per hour for deposition or trial. I have billed approximately 42,500 dollars through this report. Payment is owed regardless of outcome and does not vary with the deduction adopted.

The appended fictional résumé lists six deposition appearances and three trial appearances during the preceding four years, along with publications from the last decade. My testimony has not been excluded in this exercise history. Credentials support my ability to analyze the economic decision; they do not allow me to replace the fact testimony of Dr. Nia Calder or the Court's legal instructions.

Questions presented and limits

I was asked to determine whether the Cobalt Finch proposal represented available capacity, identify work required before production, estimate incremental cost, compare its schedule with Aurora Quay, quantify delay likely avoided, and explain a net mitigation calculation. I was also asked to assess whether damages categories can be evaluated separately.

I assume for this analysis that Granite Heron caused the temperature event and that Blue Ember reasonably rejected the lot for its intended validation use. Those assumptions isolate mitigation; they are not concessions by Granite Heron in the litigation.

I do not interpret the contract, determine who bears a legal burden, or decide which issue was properly raised in a trial motion. I do not say Cobalt Finch guaranteed scientific success. My opinion is that the available course was sufficiently capable and economically favorable that a reasonable development organization would have secured and pursued it.

I do not independently value the 1.35 million dollar original lot or audit the 850,000 dollar replacement category. My calculation addresses avoidable delay. Separate defense arguments may reduce other amounts, but they are not stacked into the 1.5 million figure.

Materials and analytical approach

I reviewed the integrated shipment writings, CTX-9 chronology, lot disposition, program schedules, Blue Ember financial records, Cobalt Finch offer and calendar, preliminary gap plan, estimated transfer budget, question log, and the Aurora Quay reservation. I considered testimony of Solis, Locke, Holloway, and Calder.

I reviewed Kellan Brooks's July 7 report and supporting schedules, including his delay model and scenario table. I used his bounded 2 million dollar delay category as the universe against which avoidance is measured. Where I disagreed, I retained his source structure so the consequence of the disagreement remains visible.

My method compares the actual replacement path with the supported earlier alternative at the decision date. It identifies common costs, incremental alternative expense, schedule milestones, execution risk, and the portion of claimed loss linked to timing. I then tests whether uncertainty materially changes the conclusion through scenarios.

I did not use later success at another Cobalt Finch project as proof BE-214 would have succeeded. I considered post-deadline evidence only where it explains records that existed by October 4, such as facility capability and what the proposed tasks meant.

Capacity was concrete and time-limited

Cobalt Finch held a real Durham suite opening for a campaign beginning December 2, 2024. The written proposal issued October 2 and remained available until 17:00 on October 4. Another sponsor booked the window after Blue Ember declined, corroborating that the reservation concerned actual scarce capacity.

The offer went beyond a generic expression of interest. It identified a suite, target date, transfer sequence, commercial structure, preliminary budget, readiness gate, staffing concept, and steps needed from Blue Ember. Acceptance would have reserved capacity and started work on October 7.

The deadline was compressed because technical work had to begin immediately and another customer was waiting. Short duration increased decision pressure but also reflected the economic value of the opportunity. Blue Ember had enough information to secure the slot subject to the disclosed readiness process.

Open capacity alone does not guarantee output. Here it was combined with relevant equipment and development-biologics experience, a defined gap plan, an estimated cost, and a calendar capable of advancing production by approximately three months. In my field, that combination is a practicable alternative rather than speculation.

Facility and process fit

Calder identified no known incompatibility that would prevent BE-214 production. Cobalt Finch had an appropriate scale range, single-use processing capability, frozen receiving and storage, trained staff, analytical options, and a quality system designed for development campaigns.

BE-214 had not previously run there. That fact created transfer work, not a conclusion of infeasibility. The preliminary fit assessment flagged equipment equivalence, mixing details, analytical allocation, batch-record adaptation, materials, and training. Those are recognizable tasks in a site transfer.

Blue Ember's own diligence did not uncover a fatal deficiency during the offer window. Its questions concerned how and when open items would be resolved. Cobalt Finch proposed a focused on-site review, staged documents, an engineering confirmation, and a joint November 22 readiness gate.

An established site such as Aurora Quay offered greater execution confidence. Economic reasonableness, however, does not require selecting only the lowest-risk path when a modest-cost alternative offers a large schedule benefit. The comparison must weigh likelihood, downside, cost, and loss threatened. The Cobalt Finch expected benefit substantially exceeded the incremental commitment.

Transfer schedule feasibility

The proposed path began with an October 7 kickoff and gap assessment through October 11. On-site review was planned for October 14 and 15. Document mapping, method responsibilities, materials, and training would proceed through early November, followed by engineering confirmation from November 11 through 15.

The parties would make a final readiness decision by November 22. Production was scheduled to start December 2, with preliminary campaign completion near December 20 and testing and record review thereafter. The model does not assume usable material instantly at manufacturing start.

I mapped the tasks against ordinary dependency and resource constraints described in Calder's records. Several workstreams could operate concurrently without eliminating a critical safeguard. Blue Ember controlled the speed of document transfer and staff availability; Cobalt Finch controlled suite preparation and its quality review.

The schedule was aggressive but credible. I find a materially greater likelihood of timely execution than of total failure or a delay extending to the March 3 Aurora Quay start. That conclusion rests on the defined task plan and known capability, not on a promise that no deviation could occur.

Decision analysis at October 4

A proper mitigation assessment uses information known when Blue Ember had to choose. By October 4, it knew the magnitude of threatened delay, had a held December slot, possessed a preliminary fit assessment, understood major transfer tasks, and had a 300,000-dollar incremental estimate.

Blue Ember also knew that Cobalt Finch was not yet program-qualified, that detailed review remained, and that some commitment could be lost if readiness failed. Those concerns were genuine. They should be measured against the expected 1.8 million dollars of delay avoided and the ability to retain contingency planning for Aurora Quay.

Blue Ember did not complete an on-site review or obtain a quantified risk assessment before declining. It favored the established site largely because of execution confidence. In my opinion, a reasonable organization facing this benefit-cost ratio would reserve the slot and promptly complete the planned diligence rather than surrender the capacity.

The decision did not require unsafe release or skipping change control. The proposal expressly retained readiness conditions. If a critical gap emerged, Blue Ember could stop, incur defined sunk cost, and return to its later path. That downside was far smaller than the time-related loss at risk.

Incremental alternative cost

Cobalt Finch estimated 300,000 dollars for transfer management, focused facility qualification, document adaptation, analytical planning, training, engineering confirmation, and the premium associated with expedited capacity. The estimate excluded ordinary replacement manufacturing that Blue Ember would pay under either site path.

I reviewed the task-level support and compared it with fictional industry ranges for development transfers of similar scale. The estimate is reasonable and includes a contingency for ordinary scope clarification. A major unanticipated issue might cost more, but no known issue by October 4 indicated that result.

Cancellation exposure was staged. Blue Ember would not lose the entire amount on acceptance day. Work completed and capacity reserved would become payable as milestones occurred. In an unsuccessful scenario, some cost would be sunk, but early diligence would limit exposure.

For the successful alternative, 300,000 dollars is the appropriate incremental cost deducted from loss avoided. It should not be added to actual damages because Blue Ember did not incur it. It should not be subtracted twice as both an expense and a risk adjustment. My calculation identifies it in one line.

Derivation of 1.8 million dollars avoided

Brooks attributes 2 million dollars to the actual timing shift. I mapped the December manufacturing path through testing and milestone effects rather than assuming all delay disappeared on December 2. The earlier course would have eliminated approximately ninety percent of the claimed time-related loss.

The remaining 200,000 dollars reflects disruption that would persist even with prompt mitigation: the September investigation, transfer mobilization, certain unavoidable schedule friction, and residual timing before replacement disposition. Thus gross delay avoided is 1.8 million dollars, not the full 2 million.

The calculation uses Blue Ember's own risk-adjusted contribution structure. I did not pair an optimistic alternative with an unadjusted commercial forecast. Milestones are advanced only where the proposed schedule affects them, and later unrelated program uncertainty remains in the model.

A catastrophic transfer-failure scenario would avoid less, but the defined capability and task evidence make it an insufficient basis to treat the opportunity as worthless. Sensitivity testing shows that even substantial reduction in expected schedule benefit leaves the decision economically favorable relative to the 300,000-dollar incremental cost.

Net mitigation deduction

The net calculation is 1.8 million dollars of avoidable delay minus 300,000 dollars required to obtain the alternative. The difference is 1.5 million dollars. Applied to the conditional 4.2 million dollar gross figure, the resulting loss is 2.7 million dollars.

The deduction concerns only avoidable delay. The original 1.35 million dollar lot cost and the 850,000 dollar replacement subtotal remain in the conditional result. That structure prevents the alternative expense from erasing an unrelated direct category.

My opinion treats the transfer cost as necessary even if some individual task later proved unnecessary. Using the full estimate is conservative for Blue Ember. It also leaves 200,000 dollars of the claimed delay in place rather than assuming perfect schedule recovery.

Granite Heron disputes damages beyond this mitigation issue. If the factfinder independently lowers gross delay below 1.8 million, the net deduction cannot exceed the supported avoidable portion. If it rejects one direct component, that change should occur in a different calculation line. The table is designed to reflect those findings without an overlapping reduction.

Response to Brooks on uncertainty

Brooks emphasizes that no contemporaneous numerical success probability existed. Operations decisions commonly proceed without a single formal probability. Evidence of capability, task completion, known gaps, schedule margin, and downside allows a professional assessment even when the sponsor did not write a percentage in real time.

His approach effectively assigns too much weight to the possibility of failure and too little to the value of securing scarce capacity. Blue Ember could have managed uncertainty through staged diligence and the November readiness gate. Choosing not to pursue the offer transformed manageable risk into the certainty of waiting for March.

Brooks correctly states that probability should not be counted twice. My 1.8 million avoidance reflects residual schedule effect and uses the same risk-adjusted economic base as his model. I then subtract the entire 300,000-dollar incremental estimate. I do not add a second generic risk penalty.

He also correctly separates direct from delay categories. Our disagreement is therefore bounded: whether the Cobalt Finch evidence supports the earlier path strongly enough to require a substantial net reduction. In my opinion, the combined facts support 1.5 million dollars.

Sensitivities and decision threshold

If gross avoidance were 1.8 million dollars and incremental cost 300,000 dollars, net benefit is 1.5 million. If avoidance were reduced to 1.2 million, the net benefit would still be 900,000. At 900,000 of avoidance, net benefit would be 600,000.

The financial break-even point is 300,000 dollars of gross avoidance, before considering nonfinancial program value. Granite Heron's supported schedule evidence substantially exceeds that threshold. Even a cautious sponsor could use staged commitment to limit exposure while preserving the higher-value opportunity.

A 300,000-dollar verdict deduction would imply either only 600,000 dollars of gross avoidance after cost or another combination with the same net result. That is arithmetically possible, but it assigns markedly less schedule benefit than the Cobalt Finch calendar and Blue Ember's own delay model support.

I do not instruct the factfinder which scenario the law permits. I show why the full net-benefit opinion is robust to ordinary uncertainty. A scenario based on total failure should be weighted against the known capability and proposed safeguards, not assumed merely because no guarantee existed.

Separate issues and nonduplication

The shipment's temperature chronology addresses performance and causation. Blue Ember's quality disposition addresses whether the lot served its validation purpose. The cost schedules measure gross loss. Cobalt Finch evidence concerns what Blue Ember could reasonably do after rejection. These questions interact sequentially but are not the same issue.

An economic deduction does not revise the logger curve, excuse the power transition, or require the original lot to be released. Conversely, proof of a carrier breach does not answer whether a later option would reduce damages. The model keeps those determinations distinct.

The direct subtotal and the timing subtotal also remain separate. A reduction in delay does not revive the rejected asset. A credit for reused replacement work should affect direct cost and should not be counted again as mitigation. Insurance or vendor reimbursement should offset only the corresponding compensated item.

This modular structure allows a verdict form to identify gross damages and mitigation expressly. It also permits later review of one component without guessing whether a jury compromised liability, causation, direct cost, delay, or post-event conduct.

Assumptions and limitations

I relied on produced records and testimony without conducting a financial-statement audit. Cobalt Finch's technical descriptions are assumed accurate within Calder's stated limits. I did not inspect real equipment because every facility in this exercise is fictional.

My conclusion assumes Blue Ember could provide the documents, materials, and staff contemplated by the transfer plan. The record identifies no preexisting impossibility, but undisclosed evidence of a critical shortage could change feasibility. A supported cost increase or schedule slippage could also change the net amount.

I accept Blue Ember's rejection and gross delay framework conditionally. If the factfinder concludes there was no caused delay, there is nothing in that category to mitigate. If it accepts only a smaller delay amount, a deduction is capped by that amount after appropriate incremental cost.

I express no view about evidentiary admissibility, contract interpretation, jury instructions, motion practice, or appellate remedy. My professional opinion concerns the economic reasonableness and quantified effect of the alternative. I will supplement if materially new information arrives and will identify any change rather than silently substituting an updated premise.

Final opinions and signature

First, Cobalt Finch offered genuine, scarce capacity for a December 2, 2024 BE-214 campaign and held it through October 4 at 17:00. Second, its facility had relevant capability and a concrete accelerated transfer plan, although program-specific approval and execution remained.

Third, a reasonable development organization would have secured and pursued the opportunity because the supported schedule benefit materially exceeded staged cost and manageable risk. Fourth, successful use of that path would have avoided 1.8 million dollars of Blue Ember's claimed 2 million delay category.

Fifth, reasonable incremental transfer and qualification cost was 300,000 dollars. The resulting net avoidable loss is 1.5 million dollars. Sixth, applying that amount to a conditional gross loss of 4.2 million yields 2.7 million dollars, without altering the 2.2 million direct subtotal.

The fictional appendices contain my source list, qualifications, billing information, prior testimony, capacity comparison, transfer network, cost schedule, avoided-delay bridge, and sensitivities. My workpapers retain the formulas underlying each table.

Dr. Elodie Park

Fictional operations-economics expert

Dated August 4, 2025

All credentials, publications, assignments, companies, sums, and events in this report were created exclusively for the synthetic training case.